

ModalAI VOXL2 ROS 2 Target Tracking Demo

Overview

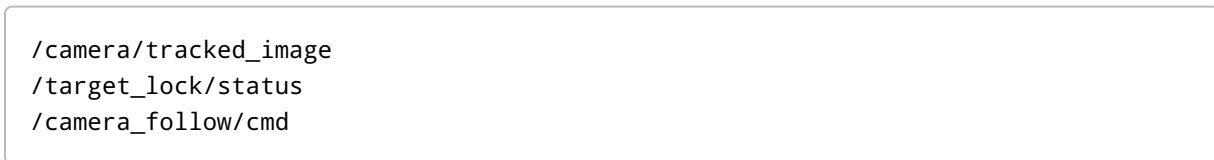
This project demonstrates manual target selection and visual tracking on a ModalAI VOXL2 platform using ROS 2, OpenCV CSRT tracking, and Foxglove visualization.

The user selects a target once and the tracker follows it without requiring YOLO, TensorRT, or object detection models.

System Architecture



Foxglove displays:



Hardware Requirements

Supported Platforms

VOXL2

Recommended minimum platform.

VOXL2 Mini

Supported for CSRT tracking.

VOXL2 + Sentinel

Recommended if future YOLO/TensorRT support is desired.

Jetson + ModalAI Flight Stack

Best performance for AI detection.

Software Requirements

VOXL Services

Verify:

```
vox1-inspect-services
```

Expected:

```
vox1-camera-server  
vox1-portal  
vox1-vision-hub
```

Check camera operation:

```
vox1-inspect-cam
```

ROS 2 Installation

Install ROS 2 support:

```
apt-get update  
apt-get install -y voxl-mpa-to-ros2
```

Launch bridge:

```
source /opt/ros/foxy/setup.bash  
  
ros2 run voxl_mpa_to_ros2 voxl_mpa_to_ros2
```

Check topics:

```
ros2 topic list
```

Expected camera topic:

```
/hires/image_raw
```

or

```
/tracking/image_raw
```

depending on VOXL configuration.

Verify frame rate:

```
ros2 topic hz /hires/image_raw
```

Docker Image Build

Create image:

```
docker build -t target-follow-demo:latest .
```

Verify:

```
docker images
```

Docker Compose

Create:

```
services:

  foxglove:
    image: target-follow-demo:latest
    container_name: foxglove_bridge

    network_mode: host
    ipc: host

    environment:
      - ROS_DOMAIN_ID=42

    command: >
      bash -c "
        source /opt/ros/humble/setup.bash &&
        source /ros2_ws/install/setup.bash &&
        ros2 launch foxglove_bridge foxglove_bridge_launch.xml port:=8765
      "

  tracker:
    image: target-follow-demo:latest
    container_name: target_tracker

    network_mode: host
    ipc: host
    privileged: true

    environment:
      - DISPLAY=${DISPLAY}
      - QT_X11_NO_MITSHM=1
      - ROS_DOMAIN_ID=42
```

```
volumes:
  - /tmp/.X11-unix:/tmp/.X11-unix
  - /dev:/dev

command: >
  bash -c "
  source /opt/ros/humble/setup.bash &&
  source /ros2_ws/install/setup.bash &&
  ros2 run target_follow_demo target_follow_node
  --ros-args
  -p image_topic:=/hires/image_raw
  "
```

Launch

Start all containers:

```
docker compose up -d
```

Check:

```
docker ps
```

Expected:

```
foxglove_bridge
target_tracker
```

Selecting a Target

The tracker uses OpenCV ROI selection.

Display access:

```
xhost +local:docker
```

The tracker window appears.

Select:

1. Click and drag box
2. Press ENTER
3. Tracker locks target

Controls:

```
ENTER = Confirm  
r      = Reselect  
q      = Quit
```

Verify Tracking

Check status:

```
ros2 topic echo /target_lock/status
```

Expected:

```
LOCKED  
TRACKING
```

Check output image:

```
ros2 topic hz /camera/tracked_image
```

Expected:

```
8-30 FPS
```

Foxglove Setup

Start bridge:

```
ros2 launch foxglove_bridge foxglove_bridge_launch.xml
```

Verify:

```
ss -lntp | grep 8765
```

Expected:

```
0.0.0.0:8765
```

Connect Foxglove

Open Foxglove Desktop.

Add connection:

```
WebSocket
```

URL:

```
ws://VOXL_IP:8765
```

Example:

```
ws://192.168.8.1:8765
```

Foxglove Layout

Image Panel

Topic:

```
/camera/tracked_image
```

Displays:

Video
Bounding Box
Tracking Center

Status Panel

Topic:

/target_lock/status

Displays:

LOCKED
TRACKING
LOST

Command Panel

Topic:

/camera_follow/cmd

Displays gimbal control output.

Gimbal Integration

The tracker publishes:

/camera_follow/cmd

Type:


```
geometry_msgs/Twist
```

Map:

```
angular.z → pan  
angular.y → tilt
```

Connect to:

```
VOXL gimbal node  
PX4 offboard controller  
Servo controller
```

IMU Integration (Optional)

If a CH341 USB IMU is attached:

```
ls /dev/ttyCH341USB0
```

Create alias:

```
ln -sf /dev/ttyCH341USB0 /dev/imu_usb
```

Run:

```
ros2 run wit_ros2_imu wit_ros2_imu
```

Verify:

```
ros2 topic echo /imu/data_raw --once
```

Performance Estimates

VOXL2

Resolution	FPS
640x480	40-60
720p	20-40
1080p	10-20

Jetson Orin

Resolution	FPS
720p	60+
1080p	30-60

Future Upgrades

The system can later be upgraded to:

YOLOv8
YOLO-NAS
TensorRT
Object Re-Identification
Multi-target tracking

without changing the Foxglove visualization architecture.